# Maintenance work order: Clean main heater E-201

**Status: DRAFT for engineering review, not approved.** Generated by PlantAgent from the workspace record; every number carries its source.

## Equipment and task

| Field | Value | Source |
|---|---|---|
| Plant | Vacuum-salt plant with vapour recompression evaporator (US 5,366,514 flowsheet) | cases/salt_mvr/case.json |
| Equipment | E-201 | P&ID |
| Class | TubularHeatExchanger | P&ID E-201 |
| Task | Clean main heater E-201 | answer 6 (muse-spark-1.3-contributor) |
| Why | Fouled heater needs 111.3% of area at 120%; clean needs 89.0% per runs vrc_120_check and vrc_120_K202_clean. | answer 6 (muse-spark-1.3-contributor) |
| Cost class | <$100k | answer 6 (muse-spark-1.3-contributor) |
| Verified in the simulator | yes | answer 6 (muse-spark-1.3-contributor) |

## Changes this work makes to the model

| Change | Source |
|---|---|
| V-201 evaporator heat-transfer coefficient: 2000 → 2500 W/m²K | answer 6 (muse-spark-1.3-contributor) |

## Isolation points (P&ID)

| Line | Connects | Valves to close and lock | Source |
|---|---|---|---|
| 2002 | in from P-201 | no valve on this line: isolate at the next item / blind | P&ID line 2002 |
| 2003 | out to V-201 | no valve on this line: isolate at the next item / blind | P&ID line 2003 |
| 4001 | out to E-102 | no valve on this line: isolate at the next item / blind | P&ID line 4001 |
| 3003 | in from D-201 | 3003/C1(GateValve) | P&ID line 3003 |

## Relief while isolated (P&ID relief scan)

| Side | Status | Detail | Source |
|---|---|---|---|
| SL | protected | relief 7001/PSV-201 with no closable valve in between | P&ID relief scan |
| ST | isolatable | relief 7002/PSV-202 only through 3003/C1: if one is closed the side has no relief | P&ID relief scan |

## Acceptance check (from the answer's field checks)

| Check | Where / how | What it changes | Source |
|---|---|---|---|
| E-201 true clean U and current fouled U | Measure E-201 steam pressure/condensing T, brine inlet/outlet T and circulation flow at steady load to back-calculate U | If clean U is below 2500, heater stays above 100% and larger clean or extra area needed | answer 6 (muse-spark-1.3-contributor) |

## Expected effect (verified proposal)

| Quantity | Before | After | Source |
|---|---|---|---|
| V-201 boil-up duty vs heat from the heating steam (2 % closure tolerance) | 99.4 % | 99.4 % | answer 6 (muse-spark-1.3-contributor) calculation record at 120 % |
| V-201 heater area needed (BPE included) vs installed | 111.3 % | 89 % | answer 6 (muse-spark-1.3-contributor) calculation record at 120 % |
| plant capacity | 100.1 % | 100.1 % | answer 6 (muse-spark-1.3-contributor) calculation record |

## Sign-off

| Role | Name | Signature | Date |
|---|---|---|---|
| Maintenance planner | | | |
| Operations (isolation) | | | |
| Process engineer (acceptance) | | | |